Chapter 11: Brain Development and Plasticity

Comprehensive Neuroscience Reference

Abstract

The brain develops from a simple sheet of neural tissue into one of nature's most complex structures, then retains remarkable capacity for reorganization throughout life. This chapter traces the major developmental milestones: neural tube formation, proliferation of neural progenitors (∼250,000 neurons/minute at peak), radial and tangential migration, axon guidance, and synaptic overproduction followed by activity-dependent pruning. We examine critical and sensitive periods for experience-dependent circuit refinement (visual plasticity, language acquisition, social attachment), the mechanisms closing these periods (perineuronal nets, myelin brakes, molecular inhibitors), and evidence that these brakes can be experimentally reopened. Adult neuroplasticity is examined through cortical map reorganization after training or sensory loss, stroke recovery mechanisms, and continuing neurogenesis. We review epigenetic mechanisms linking early experience to lifelong brain function and psychiatric risk. Finally, we address what makes the human brain unique: exceptional size, unique cell types (von Economo neurons, expanded outer radial glia), expanded long-range connectivity, and key genes (FOXP2, ASPM, ARHGAP11B, SRGAP2C) that underwent positive selection in our lineage.

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1. Prenatal Brain Development

1.1 Neural Induction and Tube Formation (Weeks 3–4)

The nervous system begins as the **neural plate** — a thickening of dorsal ectoderm induced by signals from the underlying notochord and mesoderm (primarily BMP inhibitors — noggin, chordin, follistatin — from the Spemann-Mangel organizer). The neural plate folds to form the **neural tube** (neurulation, ∼week 4), the precursor to the entire CNS:

- Anterior neural tube → brain (three primary vesicles: prosencephalon, mesencephalon, rhombencephalon)
- Posterior neural tube → spinal cord
- Neural crest cells (from neural folds) → peripheral nervous system, melanocytes, craniofacial structures

Neural tube defects (failure of neural tube closure): Anencephaly (anterior), spina bifida (posterior). Risk reduced ∼70% by maternal folate supplementation.

Patterning: Morphogen gradients establish brain axes:

- **Anterior-posterior**: Wnt, FGF, retinoic acid gradients (low Wnt → forebrain; high Wnt → hindbrain)
- **Dorsal-ventral**: BMP (dorsal) vs. Sonic hedgehog/Shh (ventral, from floor plate and notochord). In the spinal cord: Shh specifies motor neurons ventrally; BMP specifies sensory interneurons dorsally.
- **Hox genes**: Specify rhombomere identity in the hindbrain (segmental organization of cranial nerve nuclei)

1.2 Proliferation (Weeks 5–25)

Ventricular zone (VZ): Neuroepithelial cells and radial glial cells undergo symmetric division (expanding the progenitor pool) and then asymmetric division (one progenitor + one neuron or intermediate progenitor). The **subventricular zone (SVZ)** — particularly expanded in humans — contains intermediate progenitor cells (IPs) and outer radial glia (oRG, a cell type vastly expanded in primates) that enable the massive cortical expansion unique to humans.

Peak neurogenesis: ∼250,000 new neurons per minute during peak proliferation. Virtually all cortical neurons are born by gestational week 25; microglia colonize the brain from yolk sac-derived precursors starting at week 4–5.

1.3 Neuronal Migration (Weeks 8–24)

Newborn neurons must travel from the proliferative zones to their final positions:

Radial migration: The dominant mode for cortical excitatory neurons. Neurons migrate along radial glial fibers from VZ/SVZ to the cortical plate. The cortex develops in an **inside-out pattern**: early-born neurons form deep layers (V, VI); later-born neurons migrate past them to form superficial layers (IV, III, II). Key molecules: Reelin (secreted by Cajal-Retzius

cells in the marginal zone — provides a stop signal for migrating neurons; reelin mutations → lissencephaly in mice, heterotopia in humans).

Tangential migration: Cortical GABAergic interneurons are born in the ganglionic eminences (subcortical proliferative zones — medial ganglionic eminence → PV+ and SST+ interneurons; caudal ganglionic eminence → VIP+ and other interneuron subtypes) and migrate tangentially through the developing cortex before radially settling into their final layers.

Migration disorders: Lissencephaly (smooth brain — absent gyri; LIS1/DCX mutations), periventricular heterotopia (neurons fail to migrate from VZ; filamin-A mutations → epilepsy), polymicrogyria, focal cortical dysplasia → epilepsy, intellectual disability.

1.4 Axon Guidance (Weeks 8–28)

Growing axons navigate to their targets via the **growth cone** — a motile structure at the axon tip containing filopodia and lamellipodia that sense guidance cues:

Molecular guidance cues:

- **Netrins**: Attract (via DCC receptors) or repel (via UNC5 receptors) at the midline (floor plate secretes netrin-1)
- **Slits**: Repulsive midline signals (bind Robo receptors) — prevent re-crossing after midline decussation
- **Semaphorins**: Primarily repulsive (Plexin/Neuropilin receptors); some attractive
- **Ephrins**: Contact-mediated; bidirectional signaling (EphA/ephrin-A = repulsion; EphB/ephrin-B = bidirectional). Establish topographic maps (e.g., retinotopic map in superior colliculus).

1.5 Synaptogenesis and Synaptic Overproduction

Synaptogenesis begins prenatally (∼week 20) but explodes postnatally:

- **Peak synapse density**: ∼150% of adult levels, reached at different ages for different cortical areas:
  - Visual cortex: Peak at ∼8 months postnatal
  - Auditory cortex: Peak at ∼3 months
  - Prefrontal cortex: Peak at ∼3.5 years
- At peak, the infant cortex contains an estimated **1,000 trillion synapses** (more than the adult brain)
- Synaptogenesis is driven by both intrinsic developmental programs and early experience

2. Postnatal Development

2.1 Synaptic Pruning

After the period of exuberant synaptogenesis, ∼40–50% of synapses are eliminated through activity-dependent pruning — the principle of "use it or lose it":

Mechanisms:

- **Activity-dependent competition**: Synapses that are correlated with postsynaptic activation are strengthened (LTP); those that are not are weakened (LTD) and eliminated
- **Complement-mediated pruning**: Microglia selectively eliminate weak synapses tagged with complement proteins (C1q → C3 → recognized by CR3 receptors on microglia → phagocytosis). Astrocytes also engulf synapses via MEGF10 and MERTK pathways.
- **Timing by brain region**:
  - Sensory cortices: Pruning largely complete by early childhood (∼5–7 years)
  - Association cortices: Continues through adolescence
  - **Prefrontal cortex**: Pruning continues into the **mid-20s** — the last cortical region to mature

Adolescent pruning: The teenage brain undergoes massive synaptic reorganization, particularly in PFC. This contributes to enhanced cognitive efficiency but also creates a window of vulnerability — aberrant pruning (excessive or insufficient) is implicated in the adolescent onset of schizophrenia (excessive pruning hypothesis — Feinberg, 1982; supported by complement C4 gene overexpression findings — Sekar et al., 2016).

2.2 Myelination

Myelination follows a predictable spatiotemporal pattern:

- **Earliest**: Brainstem and cerebellar peduncles (prenatal)
- **Early postnatal**: Sensory and motor pathways, internal capsule
- **Childhood**: Association areas, corpus callosum
- **Latest**: Prefrontal cortex — not complete until **age 25–30**

The late myelination of PFC explains the protracted development of executive function, impulse control, and decision-making in adolescence. Myelination is activity-dependent — experience (learning, practice) promotes oligodendrocyte differentiation and myelin formation on active axons (adaptive myelination).

2.3 Critical and Sensitive Periods

Critical periods are developmentally restricted time windows during which experience is necessary for normal circuit maturation. After closure, the circuit becomes resistant to experience-dependent modification.

Ocular dominance plasticity (classic model): Monocular deprivation during the visual critical period (∼1–5 years in humans; peak ∼2–3 months in kittens) shifts cortical ocular dominance toward the open eye → amblyopia in the deprived eye. Mechanisms of critical period regulation:

1. **Opening**: Maturation of PV+ fast-spiking interneurons and GABAergic inhibition (enhancing signal-to-noise ratio)
2. **Closure**: Molecular "brakes" — perineuronal nets (PNNs, extracellular matrix structures ensheathing PV+ cells), myelin-associated inhibitors (Nogo receptor signaling), CREB/HDAC-dependent transcriptional changes, Lynx1 (endogenous nicotinic receptor modulator)
3. **Reopening** (experimentally): Removing brakes — chondroitinase ABC (dissolves PNNs), dark rearing, fluoxetine (reduces intracortical inhibition), valproate (HDAC inhibitor → chromatin remodeling) — can reopen plasticity in adulthood. Clinical implications for adult amblyopia treatment.

Language critical period: Native language phoneme discrimination is sharpened during the first year (loss of non-native phoneme distinction by ∼10–12 months). Grammar acquisition has a sensitive period extending to ∼puberty (Lenneberg, 1967). Second language proficiency (especially accent) declines with age of acquisition, though adult language learning remains possible (attenuated sensitive period, not strict critical period).

Attachment/Social: Early social deprivation (Romanian orphan studies) causes lasting deficits in stress regulation (HPA axis), social behavior, and brain development (reduced white matter, altered amygdala reactivity). Sensitive period for secure attachment formation extends through first ∼2 years.

3. Adult Neuroplasticity

The adult brain retains substantial capacity for structural and functional reorganization:

3.1 Synaptic Plasticity

- LTP and LTD continue throughout life (see Chapter 7)
- Experience-dependent spine formation and elimination occur in adult cortex (demonstrated by in vivo two-photon imaging in rodents); ∼5–10% of cortical spines turn over monthly in adults
- Learning and environmental enrichment increase spine density and dendritic complexity

3.2 Cortical Map Plasticity

Use-dependent expansion: Cortical representations expand with intensive training:

- String instrument players: Expanded cortical representation of the left hand fingers in S1
- Braille readers: Expanded representation of the reading finger; visual cortex recruited for tactile processing
- London taxi drivers: Enlarged posterior hippocampus (spatial navigation)

Deprivation-driven reorganization: After sensory loss:

- **Amputation**: The cortical region formerly representing the lost limb is "invaded" by representations of neighboring body parts. May contribute to phantom limb pain (Ra-

machandran's remapping hypothesis; though more recent evidence suggests the original representation persists but becomes noisy rather than being replaced).

- **Blindness**: Visual cortex is recruited for auditory, tactile, and linguistic processing (cross-modal plasticity). Early blind individuals show V1 activation during Braille reading, verb generation, and sound localization — and TMS to V1 disrupts these abilities, demonstrating functional relevance.
- **Deafness**: Auditory cortex recruited for visual processing, especially peripheral visual attention and sign language.

3.3 Stroke Recovery and Rehabilitation

After stroke, recovery involves multiple plastic mechanisms:

1. **Acute (days–weeks)**: Resolution of edema and diaschisis (dysfunction of remote connected regions)
2. **Subacute (weeks–months)**: Perilesional reorganization — surviving tissue near the lesion assumes functions of damaged tissue; unmasking of latent connections; axonal sprouting
3. **Chronic**: Contralesional reorganization — homologous areas in the opposite hemisphere may compensate (though excessive contralesional activity can sometimes be maladaptive — interhemispheric inhibition imbalance)

Constraint-induced movement therapy (CIMT): Restraining the intact limb forces use of the affected limb → drives use-dependent cortical reorganization → improved function. Demonstrates that adult cortical maps remain dynamic and responsive to behavioral demand.

3.4 Adult Neurogenesis

See Chapter 2 for details on hippocampal (SGZ) and olfactory (SVZ) neurogenesis.

4. Epigenetics and Brain Development

Epigenetic mechanisms regulate gene expression without altering DNA sequence, providing a molecular interface between genes and environment:

4.1 DNA Methylation

Addition of methyl groups to cytosine (typically at CpG dinucleotides) by DNA methyltransferases (DNMTs) → generally suppresses gene transcription. In neurons:

- Activity-dependent changes in DNA methylation regulate synaptic plasticity genes
- BDNF promoter methylation: Childhood adversity → increased BDNF promoter methylation → reduced BDNF expression → associated with increased depression risk
- FKBP5 demethylation: Childhood trauma + FKBP5 risk allele → epigenetic changes in glucocorticoid receptor regulation → lifelong HPA axis dysregulation

4.2 Histone Modification

Histones (protein spools around which DNA is wrapped) can be modified by:

- **Acetylation** (by HATs): Opens chromatin → promotes transcription. HDAC inhibitors enhance memory consolidation and critical period plasticity.
- **Methylation** (by HMTs): Can activate or repress depending on the specific residue (H3K4me3 = activation; H3K9me3 = repression)

4.3 Non-coding RNAs

- **MicroRNAs (miRNAs)**: Regulate mRNA translation; involved in synaptic plasticity, neurogenesis, and neurodegeneration
- **Long non-coding RNAs (lncRNAs)**: Regulate gene expression programs during brain development; many are primate-specific and may contribute to human brain evolution

4.4 Transgenerational Epigenetic Effects

Rodent studies suggest stress-induced epigenetic changes can be transmitted to offspring (e.g., paternal stress alters offspring HPA axis via sperm miRNA changes). Evidence in humans is preliminary but includes: Dutch Hunger Winter cohort (prenatal famine exposure → altered DNA methylation detectable decades later), Holocaust survivor offspring studies (altered cortisol metabolism). The robustness and mechanisms of transgenerational epigenetic inheritance in humans remain actively debated.

5. Brain Evolution: What Makes the Human Brain Unique

Size: Human brain volume (∼1,400 cm^3) is ∼3× larger than expected for a primate of our body size (encephalization quotient ∼7). The neocortex is disproportionately expanded, especially PFC (though prefrontal proportion may not be uniquely large in absolute terms relative to other great apes — this is debated).

Unique cell types:

- Von Economo neurons (spindle neurons) in anterior cingulate and insular cortex — present only in great apes, elephants, cetaceans; possibly related to social cognition and rapid interoceptive assessment
- Expanded outer radial glia (oRG) population in the outer SVZ — enables the massive expansion of upper cortical layers (II/III) in humans

Connectivity: Human brain has disproportionately expanded long-range corticocortical white matter connections, particularly in PFC; expanded corpus callosum; more complex dendritic arbors in cortical neurons.

Genetic basis: Key genes implicated in human brain evolution:

- **FOXP2**: Transcription factor; mutations cause speech/language disorder; human-specific amino acid changes under positive selection

- **ASPM, CDK5RAP2, MCPH1**: Microcephaly genes; variants associated with brain size; under positive selection in human lineage
- **HAR1 (Human Accelerated Region 1)**: Non-coding RNA expressed in Cajal-Retzius cells during cortical development; rapidly evolved in humans
- **SRGAP2C**: Human-specific gene duplication; promotes spine density and neoteny of spine maturation
- **ARHGAP11B**: Human-specific gene; promotes basal progenitor amplification in outer SVZ → cortical expansion (when expressed in marmoset brain, causes cortical folding in a normally smooth-brained area)

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